



Smart Irrigation System

From a Tinkercad Circuit Simulation to an Interactive 3D Digital Twin



Tinkercad · Arduino Uno



Three.js · WebGL Scene



Automated Moisture Control

Project Overview

A two-part build: a working circuit prototype, and a real-time 3D visualization that brings it to life.



Sense Soil Moisture

A soil-moisture probe and TMP36 temperature sensor continuously feed live readings to the Arduino Uno.



Set a Custom Threshold

A 250 kΩ potentiometer lets the operator dial in exactly how dry the soil must get before watering starts.



Automate the Response

When soil crosses the threshold, the Arduino drives a DC pump motor, alert LEDs and a piezo buzzer automatically.



Display & Visualize

Readings appear on a 16×2 I2C LCD in the physical build — and on a live HUD inside a full 3D farm digital twin.

System Architecture

The Arduino Uno sits at the center, reading sensors and driving outputs in real time.

INPUTS



Soil Moisture Sensor

Analog probe in the bed



TMP36 Sensor

Ambient temperature



Threshold Dial

250 kΩ potentiometer



Arduino Uno R3

the control loop



OUTPUTS



DC Pump Motor

Drives irrigation



16×2 I2C LCD

On-device readout



Buzzer + LEDs

Dry-soil alerts



PART ONE

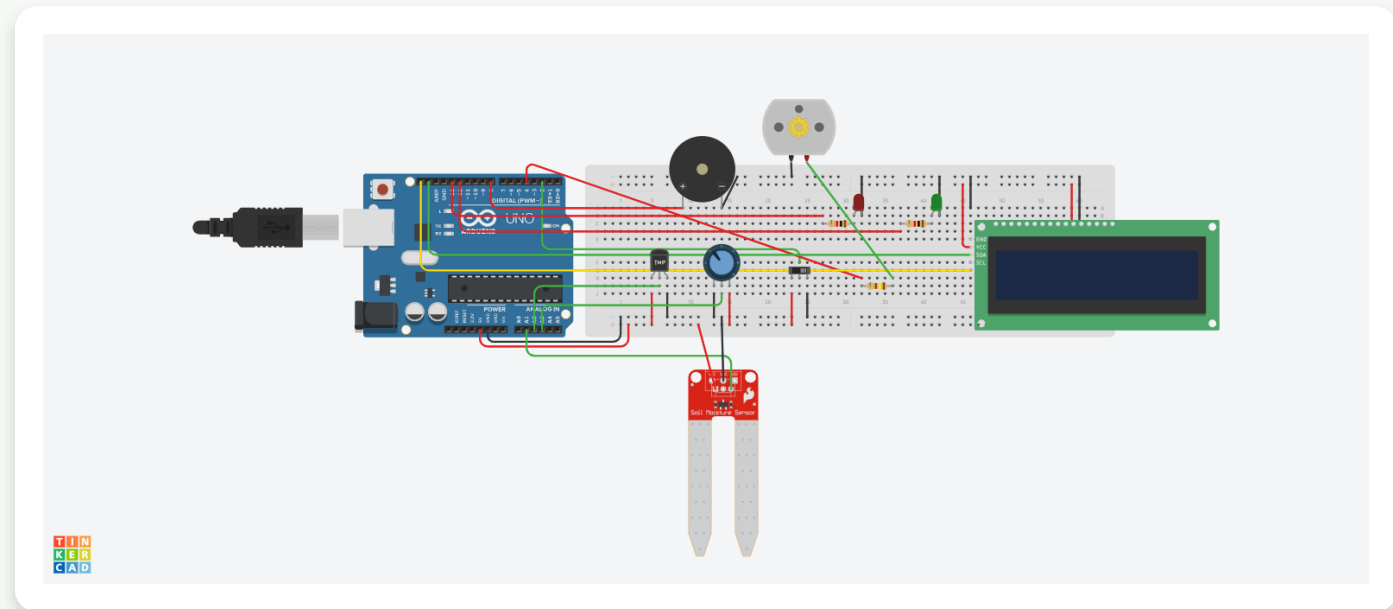
Circuit Design in Tinkercad

Prototyping the sensing and control hardware before writing a line of code.

The Circuit

Simulated end-to-end in Tinkercad before any hardware was assembled.

- **Arduino Uno R3** — control unit
- **Soil Moisture Sensor** — reads soil water content
- **TMP36 Sensor** — ambient temperature
- **DC Pump Motor** — irrigation actuator
- **16×2 I2C LCD** — on-device readout
- **250 kΩ Potentiometer** — threshold dial
- **Piezo Buzzer + LEDs** — dry-soil alerts
- **Slide Switch + Resistors** — power & current control



Full schematic — Tinkercad Circuits simulation

12

Components

1

Microcontroller

100%

Simulated First

Bill of Materials

Every component used in the Tinkercad build.

Ref	Component	Qty
D1	Red LED	1
D2	Green LED	1
U1	Arduino Uno R3	1
M1	DC Motor	1
SEN1	Soil Moisture Sensor	1
U2	Temperature Sensor (TMP36)	1
PIEZO1	Piezo Buzzer	1
—	Breadboard	1
U7	16×2 I2C LCD (MCP23008, 0x20)	1
Rpot3	250 kΩ Potentiometer	1
S1	Slide Switch	1
R3, R4	1 kΩ Resistor	2
R2	139 Ω Resistor	1

How the Circuit Works

From raw sensor signal to automatic watering, in four steps.

1



Sense

Soil-moisture and TMP36 sensors sample the bed and feed raw values to the Arduino every cycle.

2



Compare

The Arduino compares the live reading against the threshold set on the potentiometer dial.

3



Decide & Act

Below threshold: pump, red LED and buzzer switch on. Above it: the green LED stays lit and it idles.

4



Display

Moisture, temperature and system state are pushed to the 16×2 LCD in real time.



PART TWO

A 3D Digital Twin, Built in Three.js

The same irrigation logic, reimagined as an explorable, real-time farm scene in the browser.

Inside the Scene

A procedurally-textured farm — sky, soil, crops and machinery — rendered live with WebGL, zero external image assets.



Sky & Weather

Dynamic sky dome with drifting clouds and a toggleable rain system.



Water Tank & Crops

A modeled tank, windmill, shed and planted crop beds fill out the farmstead.



Control Pad Dial

A draggable in-scene dial sets the moisture threshold, mirroring the potentiometer.



Live HUD

Moisture, threshold, temperature, rain, tank and state, updated every frame.



Day / Night Cycle

Lighting, fog and sky tint shift smoothly between day and night.



Zero Image Assets

Grass, soil, concrete and wood textures are all generated on canvas at runtime.

Play With It

The scene isn't just a render — it responds to input in real time.



Live HUD Readout

Moisture, threshold, temp, rain, tank and state at a glance.



Draggable Threshold Dial

Grab the control-pad dial to retune the watering threshold live.



One-Click Dry Soil

Trigger a dry-soil event instantly and watch the response cascade.



Rain Trigger

Simulate rainfall and watch the tank and HUD react.



Day / Night Toggle

Switch lighting and mood with a single click.

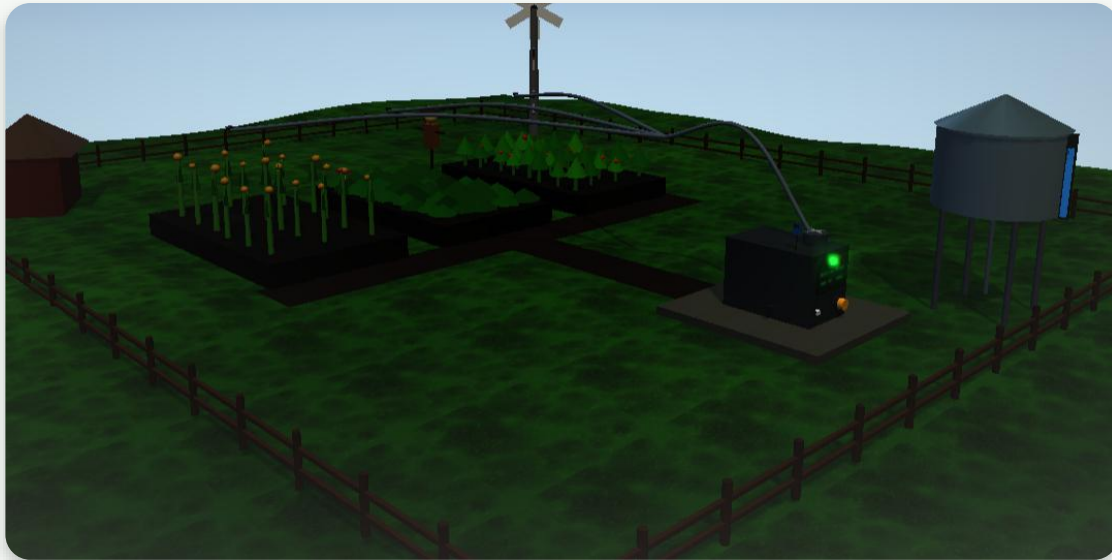


Orbit & Zoom

Drag to orbit the camera, scroll to zoom through the farm.

The Digital Twin in Action

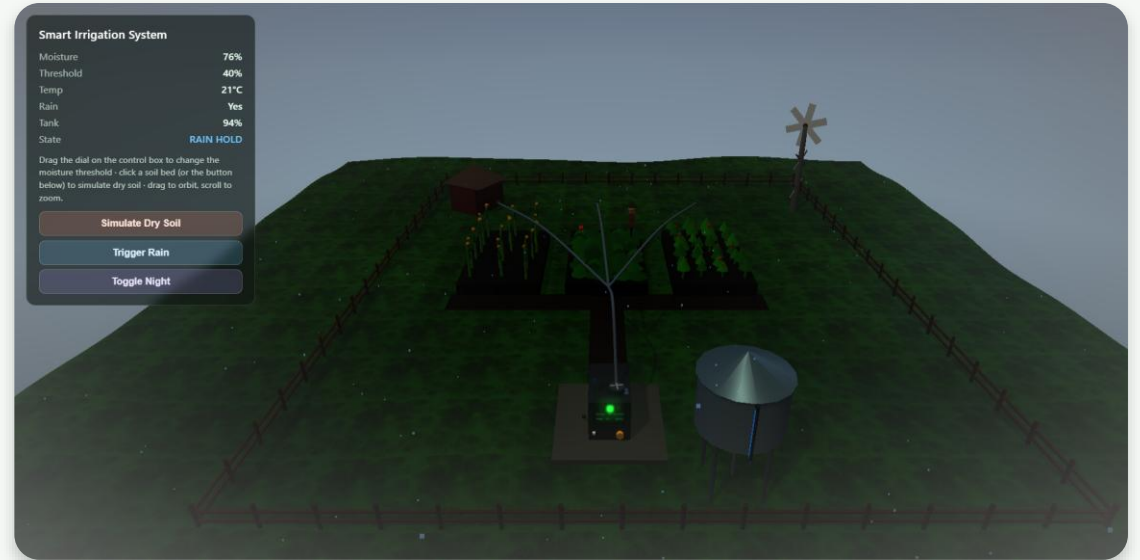
Screenshots captured live from the Three.js scene running in the browser.



IDLE

Farm view: soil moisture is above the threshold

The green status LED on the control box stays lit and the pump stays off while the soil is healthy.



LIVE HUD

Rain hold at night, with the live dashboard

Moisture 76%, threshold 40%, tank 94%: rain and night mode are on, so the system holds off pumping.

Two Builds, One System

A physical prototype and a digital twin, both modeling the same irrigation logic.



Physical Prototype

TINKERCAD

- Real Arduino Uno C++ logic on breadboard-accurate wiring
- Validates sensor thresholds and component behavior before buying hardware
- LCD, buzzer and LED feedback exactly as it runs on real hardware
- Safe place to debug circuits, with zero risk of shorts



Digital Twin

THREE.JS / WEBGL

- Full 3D farm environment rendered live in any modern browser
- Real-time HUD plus interactive threshold dial, rain and day/night triggers
- Zero build step — static HTML/JS/CSS loaded straight from a CDN import map
- Ideal for demos, presentations and exploring “what if” scenarios instantly

Impact & What's Next

What this project demonstrates — and where it can go.

IMPACT



Bridges hardware prototyping and software visualization in one project



Demonstrates a full sense → decide → act embedded control loop



Built entirely from low-cost, widely available components



The digital twin makes the system explainable to a non-technical audience

FUTURE SCOPE



Stream live sensor data from the real board into the 3D HUD



Solar-powered, field-deployable version of the circuit



Companion mobile app for remote threshold control



Multi-zone irrigation with per-bed moisture mapping

CONCLUSION

One irrigation loop, built twice

We designed a complete smart irrigation system: a Tinkercad Arduino circuit that senses soil moisture, decides against a dial-set threshold and acts automatically, and a Three.js digital twin that lets anyone watch that same logic run live in the browser.

4

step sense-to-act loop

2

builds sharing one logic

0

image files in the 3D scene



Automated, closed-loop control

Soil moisture dropping below the potentiometer threshold switches on the pump, red LED and buzzer; recovery returns the system to a green idle.



Validated before any hardware

The full circuit was simulated in Tinkercad, so wiring, thresholds and component behaviour were tested with zero risk of shorts.



Explainable to anyone

The Three.js digital twin mirrors the same logic live, with a draggable dial plus rain and day/night controls, in any modern browser.

References

Tools, platforms, datasheets and project files used to build the system.



Software & Platforms

- [1] Autodesk Tinkercad Circuits**
Circuit design and simulation
tinkercad.com/circuits
- [2] Three.js (r160)**
3D engine, OrbitControls, bloom effect
threejs.org/docs
- [3] WebGL (Khronos Group)**
Web graphics standard behind Three.js
khronos.org/webgl
- [4] jsDelivr CDN**
Serves three@0.160.0 via an import map
jsdelivr.com/package/npm/three



Hardware Docs

- [5] Arduino Uno Rev3**
Board specifications and pinout
docs.arduino.cc/hardware/uno-rev3
- [6] TMP36 Temperature Sensor**
Analog Devices datasheet
analog.com/en/products/tmp36.html
- [7] MCP23008 I/O Expander**
Microchip datasheet (I2C LCD interface)
microchip.com/en-us/product/MCP23008
- [8] Arduino LiquidCrystal Library**
16x2 LCD control reference
docs.arduino.cc/libraries/liquidcrystal



Project Files & Credits

- [9] Smart Irrigation System .pdf / .png**
Tinkercad circuit schematic export
Project folder
- [10] bom.csv**
Tinkercad bill of materials export
Project folder
- [11] index.html · style.css · main.js**
Three.js digital twin source code
Project folder
- [12] Feather Icons (MIT licence)**
Icon set used throughout this deck
feathericons.com



Thank You

Smart Irrigation System — Tinkercad Circuit & Three.js Digital Twin